

Chapter 12

Electrostatics in media

12.1 Averaging over molecular fields

We want the electric field far from the molecules of a material – obtained by averaging over space and time. A molecule at position $\vec{x}_i$ contributes charge and dipole densities

$$\rho_{\text{mol}} = \sum_i q_i \delta^3(\vec{x} - \vec{x}_i) \quad , \quad \vec{P}_{\text{mol}} = \sum_i \vec{p}_i \delta^3(\vec{x} - \vec{x}_i) \quad (12.1)$$

The potential of the i 'th molecule, expanded in multipoles, is

$$\begin{aligned} \phi_i &= \frac{q_i}{|\vec{x} - \vec{x}_i|} + \vec{p}_i \cdot \frac{\vec{x} - \vec{x}_i}{|\vec{x} - \vec{x}_i|^3} + \dots \\ &= \frac{q_i}{|\vec{x} - \vec{x}_i|} + \vec{p}_i \cdot \vec{\nabla}_i \left(\frac{1}{|\vec{x} - \vec{x}_i|} \right) \end{aligned} \quad (12.2)$$

with $\vec{E}_i = -\vec{\nabla}\phi_i$. Summing over the molecules in a cell of volume V ,

$$\vec{E}(\vec{x}) = -\vec{\nabla} \int_V dV' \left[\frac{\rho_{\text{mol}}(\vec{x}')}{|\vec{x} - \vec{x}'|} + \vec{P}_{\text{mol}}(\vec{x}') \cdot \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} \right] \quad (12.3)$$

where V is the cell volume. Define the spatial average over the cell

$$\langle \vec{E}(\vec{x}) \rangle = \frac{1}{V} \int \vec{E}(\vec{x} + \vec{x}'') dV'' \quad (12.4)$$

and write

$$\langle \rho_{\text{mol}}(\vec{x}) \rangle \equiv \rho(\vec{x}) \quad , \quad \langle \vec{P}_{\text{mol}}(\vec{x}) \rangle \equiv \vec{P}(\vec{x}) \quad (12.5)$$

Changing integration variable, the average field takes the form

$$\langle \vec{E}(\vec{x}) \rangle = -\vec{\nabla} \int dV' \left[\frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} + \vec{P}(\vec{x}') \cdot \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} \right] \quad (12.6)$$

Typically, the molecular charge density averages to zero, and any external added charge appears through

$$\rho(\vec{x}) \rightarrow \rho(\vec{x}) + \rho_{\text{free}}(\vec{x}) \quad (12.7)$$

12.2 The displacement field

Hereafter, write $\langle \vec{E}(\vec{x}) \rangle \equiv \vec{E}(\vec{x})$. Taking the divergence of (12.6)

$$\vec{\nabla} \cdot \vec{E}(\vec{x}) = - \int dV' \left[\rho(\vec{x}') \vec{\nabla}^2 \frac{1}{|\vec{x} - \vec{x}'|} + \vec{P}(\vec{x}') \cdot \vec{\nabla}' \left(\vec{\nabla}^2 \frac{1}{|\vec{x} - \vec{x}'|} \right) \right] \quad (12.8)$$

and using

$$\vec{\nabla}^2 \frac{1}{|\vec{x} - \vec{x}'|} = -4\pi \delta^3(\vec{x} - \vec{x}') \quad , \quad -4\pi \vec{\nabla}' \delta^3(\vec{x} - \vec{x}') = +4\pi \vec{\nabla} \delta^3(\vec{x} - \vec{x}') \quad (12.9)$$

we get

$$\begin{aligned} \vec{\nabla} \cdot \vec{E} &= 4\pi \int \rho(\vec{x}') \delta^3(\vec{x} - \vec{x}') dV' \\ &\quad - 4\pi \vec{\nabla} \cdot \int \vec{P}(\vec{x}') \delta^3(\vec{x} - \vec{x}') dV' \end{aligned} \quad (12.10)$$

The second integral is $\vec{P}(\vec{x})$ – the polarization of the material. Hence

$$\vec{\nabla} \cdot (\vec{E} + 4\pi \vec{P}) = 4\pi \rho \quad (12.11)$$

with ρ the free charges. Defining the displacement $\vec{D} \equiv \vec{E} + 4\pi \vec{P}$, we then have

$$\vec{\nabla} \cdot \vec{D} = 4\pi \rho \quad (\text{free charges}) \quad , \quad \vec{\nabla} \cdot \vec{P} = -\rho_p \quad (\text{polarization charges}) \quad (12.12)$$

We still have

$$\vec{\nabla} \times \vec{E} = 0 \quad (12.13)$$

12.3 Susceptibility and permittivity

$\vec{P}$ is induced by the external $\vec{E}$ through two mechanisms: dipole orientation alignment, which is temperature dependent; and molecular polarization, which is temperature independent. Often $\vec{P} \propto \vec{E}$, and we write

$$\vec{P} = \chi_e \vec{E} \quad (12.14)$$

where χ_e is the electrical susceptibility. Then

$$\vec{D} = \vec{E} + 4\pi \chi_e \vec{E} = (1 + 4\pi \chi_e) \vec{E} \equiv \epsilon \vec{E} \quad (12.15)$$

where ϵ is the permittivity, and

$$\vec{\nabla} \cdot \vec{E} = \frac{4\pi \rho}{\epsilon} \quad (12.16)$$

More generally, one has a weak- E expansion

$$P^i = a_{ij} E^j + b_{ijk} E^j E^k + \dots \quad (12.17)$$

If the material is isotropic and linear, $a_{ij} = \chi_e \delta_{ij}$.

12.4 Boundary conditions

In summary

$$\vec{\nabla} \cdot \vec{D} = 4\pi \rho \Rightarrow (\vec{D}_2 - \vec{D}_1) \cdot \hat{n} = 4\pi \sigma \quad (12.18)$$

$$\vec{\nabla} \times \vec{E} = 0 \Rightarrow (\vec{E}_2 - \vec{E}_1) \times \hat{n} = 0 \quad (12.19)$$

Note that $\vec{E} = \vec{D} - 4\pi \vec{P}$, so that

$$\begin{aligned} (\vec{E}_2 - \vec{E}_1) \cdot \hat{n} &= 4\pi \sigma - 4\pi (\vec{P}_2 - \vec{P}_1) \cdot \hat{n} \\ &= 4\pi (\sigma - \sigma_p) \equiv 4\pi \sigma_{\text{tot}} \end{aligned} \quad (12.20)$$

where $-\sigma_p$ is the excess charge on the dielectric surfaces from the polarization of the material. If there are no free charges,

$$(\vec{D}_2 - \vec{D}_1) \cdot \hat{n} = 0 \quad , \quad (\vec{E}_2 - \vec{E}_1) \times \hat{n} = 0 \quad (12.21)$$

12.5 Example: point charge near a dielectric interface

Consider a point charge q at height d above a plane interface at $z = 0$ separating two dielectrics – permittivity ϵ_1 above and ϵ_2 below, as depicted in the Figure. The potentials satisfy

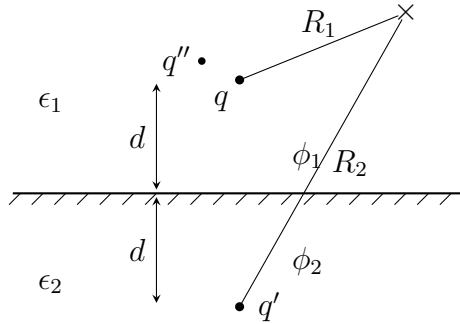


Figure 12.1: Point charge q at height d above the plane interface between dielectrics ϵ_1 (region 1, above) and ϵ_2 (region 2, below). The image charge q' sits at $z = -d$, and q'' at $z = +d$.

$$\nabla^2 \phi_1 = -\frac{4\pi \rho}{\epsilon_1} \quad , \quad \nabla^2 \phi_2 = 0 \quad (12.22)$$

and the interface is not a constant- ϕ surface. We use the method of images to find the image charge q' : write

$$\phi_1 = \frac{1}{\epsilon_1} \left(\frac{q}{R_1} + \frac{q'}{R_2} \right) \quad , \quad \phi_2 = \frac{1}{\epsilon_2} \frac{q''}{R_1} \quad (12.23)$$

with q' at $z = -d$ and q'' at $z = +d$. Using the cylindrical coordinate r ,

$$R_1 = \sqrt{r^2 + (d - z)^2} \quad , \quad R_2 = \sqrt{r^2 + (d + z)^2} \quad (12.24)$$

The boundary conditions

$$E_{1r} = E_{2r} \quad \text{at } z = 0 \quad , \quad D_{1z} = D_{2z} \quad \text{at } z = 0 \Rightarrow \epsilon_1 E_{1z} = \epsilon_2 E_{2z} \quad (12.25)$$

then give

$$q + q' = \frac{\epsilon_1}{\epsilon_2} q'' \quad , \quad q - q' = q'' \quad (12.26)$$

which lead to

$$q' = \frac{\epsilon_1 - \epsilon_2}{\epsilon_1 + \epsilon_2} q \quad , \quad q'' = \frac{2\epsilon_2}{\epsilon_1 + \epsilon_2} q \quad (12.27)$$

If $\epsilon_2 \rightarrow \infty$, region 2 is like a conductor (because $\vec{E}_2 \rightarrow 0$). The field-line patterns for the various cases are sketched in the Figure.

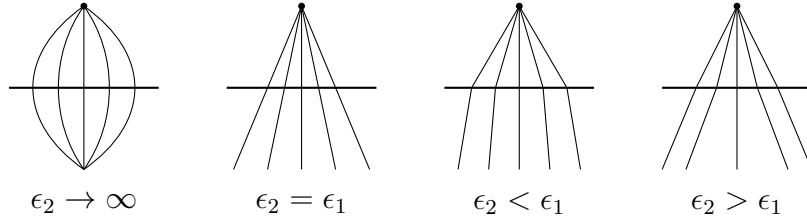


Figure 12.2: Field-line patterns of a point charge above a dielectric interface for $\epsilon_2 \rightarrow \infty$, $\epsilon_2 = \epsilon_1$, $\epsilon_2 < \epsilon_1$, and $\epsilon_2 > \epsilon_1$.

12.6 Computing polarization charges

The polarization charges follow from

$$\rho_p = -\vec{\nabla} \cdot \vec{P} \Rightarrow \sigma_p = -(\vec{P}_2 - \vec{P}_1) \cdot \hat{n} \quad (12.28)$$

computed from $\vec{P} = \chi_e \vec{E}$.